TrustShift

inance without intermediarie

Trustless P2P payments on Solana — the middleman is the program.

Superteam Poland Track — HackYeah 2026

The problem: the middleman owns your money

- Every P2P payment leans on a bank, PSP, or app that can hold, block, reverse, or freeze funds — and decides who gets served.
- Freelancers invoice strangers and wait, exposed to platform freezes and unilateral reversals.
- The trust requirement is the product. Remove it, and payments become as programmable — and as neutral — as math.

What TrustShift is

- A Solana program (devnet-first) holding the terms of a P2P transfer.
- lock — payer deposits into a program-owned escrow PDA with a deadline.
- cancel — BEFORE the deadline, the payer reclaims.
- settle — AFTER the deadline, ANYONE pushes the funds to the payee.
- No backend. No multisig service. No arbiter. ~300 lines of Rust.

The key property

- Once the deadline passes, the payer cannot block settlement.
- Nobody can. The program is the only judge.
- A third party with zero authority over the escrow can force settlement permissionlessly — proven in the demo, where a random keypair settles and the payer's key is never used again.

Target user & rationale

- Freelancers and online sellers invoicing clients they don't fully trust.
- The intermediary today: the invoicing/payment platform that can freeze the payout or side with either party arbitrarily.
- With TrustShift: the client locks funds up front — provably committed on-chain and payable at the agreed deadline.
- The client keeps a cancel window for disputes — BEFORE the deadline only.

The user flow (devnet)

- Connect wallet → lock transaction → escrow PDA visible in explorer → settle transaction → payout visible in explorer.
- The escrow account exposes payer, payee, deadline, and amount in plain account data — auditable by anyone.
- After settle, the emptied escrow and the payee's new balance are both explorer-visible.

Why it's trustless for real

- The deadline check, the settle authorization, and the fund movement all execute inside the on-chain program — not a backend.
- lock rejects past deadlines, zero amounts, duplicate escrows.
- cancel rejects strangers and late calls.
- settle rejects early calls, wrong payees, and tampered accounts.

Demo

- app/demo.py runs the full flow in-process (LiteSVM, no network):

  lock → state verify → early-settle rejected → cancel → refund →

  time-warp → permissionless third-party settle.
- Assert-based: exits non-zero on any failure. ALL CHECKS PASSED.

Real-world potential

- Escrow-as-a-primitive for freelance, marketplace, and B2B settlement.
- Composable: deadline + permissionless settle is the trust kernel other flows (milestones, conditional refunds, revenue shares) can extend.
- Zero marginal infrastructure: no servers to run or trust.

Links

- Public repo: github.com/adeviwon/trustshift
- Program: 6FoeU1wFmhh7DrWUvaeGVyFFJNx67VzS78hRjpq6Jpo8 (devnet)
- Run the demo: pip install solders litesvm && python3 app/demo.py